Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/31/2008
PI Name:	Foster, Russell Grant Ph.D.		
Project Title:	The Role and Characterization of Novel Photoreceptor Mechanisms Regulating Circadian Rhythms, Sleep, Body Temperature and Heart Rate		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	UNITED KINGDOM		
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	01/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-HPF00301		
Performance Goal No.:			
Performance Goal Text:			
	(1) The original aims of the project: The long-term aims of this project are to provide effective countermeasures for the detrimental effects of the irregular and altered patterns of light exposure in space and on the surface of other planets. These countermeasures will depend upon an understanding of the basic mechanisms of how the eye perceives light. All photoreception within the eye of mammals was considered the province of the rods and cones. However, our work has shown that varied aspects of physiology and behavior are regulated by gross changes in environmental light and that these irradiance changes are detected by non-rod, non-cone ocular photoreceptors. In 2003 we were able to demonstrate in the mouse that a small number (~ 1%) of retinal ganglion cells are directly photosensitive to light. It is these cells, termed photosensitive retinal ganglion cells (pRGCs) that seem to play a critical role in regulating varied aspects of our physiology and behavior. The central aims of the research that we have undertaken was to characterize the molecular mechanism of this unexplored		

Task Description:	photoreceptor system of the eye, and determine the extent to which the pRGCs contribute to varied aspects of physiology and behavior. In this regard our proposal spans 1-4 of the Countermeasures Readiness Levels (CRL) of NASA's Biomedical Research and Countermeasures Program. By employing a unique transgenic mouse models and by taking advantage of the new opportunities created by a range of post-genomic technologies, we addressed two broad questions: Question 1: What is the role of pRGCs in the regulation of general physiology and behavior? Question 2: What are the molecular mechanisms of non-rod, non-cone ocular photoreceptors? We have greatly advanced are knowledge in these two areas and we hope that NSBRI will take this knowledge forward for the development of drugs aimed at the manipulation of human circadian rhythms, sleep, mood and performance, and for the design of new lighting sources that are either highly effective in regulating these novel photoreceptors or leave them largely unstimulated. (2) The key findings to-date: (i) Light detection by the eye is mediated by rods, cones and photosensitive retinal ganglion cells (pRGCs). The photopigment of the pRGCs has been the subject of considerable debate. From our previous data, the melanopsin gene and its protein product emerged as the strongest candidate for this photopigment. As part of our collaborative interactions we have been able to show, using gene expression approaches, that melanopsin forms the photopigment of the pRGCs. These data were published in Nature. ii) Light detection by the eye is mediated by rods, cones and photosensitive retinal ganglion cells (pRGCs), but it has remained unclear which of these photoreceptor classes modulate sleep and alertness. Our data provide the first direct physiological and behavioural evidence that pRGCs modulate sleep. Furthermore we show that mice which have rods and cone but lack melanopsin no longer show sleep induction in response to light exposure. Thus the pRGCs provide either the dominant or exclusive photic input into the sleep switch of the brain. This work is being prepared for a submission to Science. iii.) Cardiovascular performance, blood pressure and core body temperature can be modified by changes in ambient light. However, these responses have only been poorly characterised. Our on-going studies suggest that the acute regulation of heart rate by environmental light in animal models arises from inputs from a time-dependent interaction between the classical rod and cone photoreceptor pathways and the pRGCs. iv.) To investigate the molecular mechanisms where by light is transduced into a physiological response in pRGCs we have undertaken a microarray project looking at light induced gene expression. 8 genes have been identified (Gnas, Gnb2l1, Gnaq, Prkcz, Pik3r1, Inadl, Slc9a3r1, Drd1a) that co-localise with melanopsin pRGCs. The impact of genetic ablation of one of these, protein kinase C zeta (Prkcz), was assessed. Prkcz-/- animals show attenuated phase shifting responses to light, reduced period lengthening under constant light and attenuated pupillary responses at high irradiances; as well as impaired light-induced gene expression in the suprachiasmatic nuclei (SCN). These attenuated responses are indistinguishable from the deficits observed in melanopsin knockout mice. This study is in press in Current Biology. (3) The impact of these findings on the aims of the project: When our project was originally reviewed support was enthusiastic but concerns were raised that we had been overly ambitious. We feel that over the duration of the grant we have more than fulfilled our set of goals. In brief, we have been able to show that the photic stimulation of pRGCs must be considered the primary target for the development of effective countermeasures for the detrimental effects of the irregular and altered patterns of light exposure in space and on the surface of other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Varied aspects of physiology and behaviour are regulated by gross changes in environmental light. Research leading up to this proposal has shown that in both man and laboratory rodents these irradiance changes may be detected by photoreceptors that are distinct from rod and cone pathways – the so called "photosensitive retinal ganglion cells" or pRGCs. Photoreceptive mechanisms underlying these non-visual effects of light remain poorly understood. Yet the importance of this novel photoreceptive system in our daily lives is profound. It is now clear that light exposure can influence alertness and sleep propensity and, in consequence, our performance of demanding tasks. Furthermore, light regulates the phase of internal (circadian) clocks, and thus the timing of rhythmic functions such as digestion, sleep and activity. The cost to countries of working out of phase with the natural light environment, or working under inappropriate light conditions, is substantial in terms of poor performance, insomnia, altered mood, and increased fatigue. For example, in the USA the cost of accidents attributable to human fatigue is estimated at $16 billion per year. New approaches to address the clinical problems that either result from, or are affected by, inappropriate lighting conditions (e.g. depression, insomnia, reduced alertness and circadian maladjustment associated with shift work, blindness or jet-lag) will be critically dependent upon a detailed understanding of the basic mechanisms of the non-rod, non-cone ocular photoreceptors. Thus a major benefit of the work outlined in this proposal will be to provide the mechanistic substrate for targeted drug development aimed at the manipulation of human circadian rhythms, sleep, mood and performance.
Task Progress:	When our project was originally reviewed support was enthusiastic but concerns were raised that we had been overly ambitious. We feel that we have fulfilled an impressive set of goals and have addressed all the stated objectives. Collectively, our findings provide the first physiological evidence that melanopsin-based pRGCs regulate sleep; that the acute regulation of heart rate by environmental light arise from a time-dependent interaction between the classical rod and cone photoreceptor pathways and melanopsin-based pRGCs; and we have identified unique elements of the melanopsin phototransduction pathway. In short, we have been able to show that the photic stimulation of pRGCs must be considered the primary target for the development of effective countermeasures for the detrimental effects of the irregular and altered patterns of light exposure in space and on the surface of other planets.

Bibliography Type:	Description: (Last Updated: 03/02/2017)
Articles in Peer-reviewed Journals	Lupi D, Sekaran S, Jones SL, Hankins MW, Foster RG. "Light-evoked FOS induction within the suprachiasmatic nuclei (SCN) of melanopsin knockout (Opn4-/-) mice: a developmental study." Chronobiol Int. 2006;23(1-2):167-79. PMID: 16687291 , Mar-2006
Articles in Peer-reviewed Journals	Peirson S, Foster RG. "Melanopsin: another way of signaling light." Neuron. 2006 Feb 2;49(3):331-9. PMID: 16446137 , Feb-2006
Awards	Foster RG. "Chair of Circadian Neuroscience, University of Oxford, October 2006." Oct-2006
Awards	Foster RG. "Chair of Neuroscience Theme Steering Committee, University of Oxford, October 2007." Oct-2007
Awards	Foster RG. "Chair, BBSRC Healthy Organism Strategy Panel, January 2007." Jan-2007
Awards	Foster RG. "Head of Department , Ophthalmology, University of Oxford, May 2007." May-2007
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Awards	Foster RG. "Raine Medical Research Institute Visiting Professor 2007, University of Western Australia, Perth, Australia." Mar-2007
Papers from Meeting Proceedings	Lucas RJ, Foster RG, Hankins MW, Barnard AR. "Melanopsin, photosensitive retinal ganglion cells and non-image forming responses to light." CIE Symposium on Light and Health: Non Visual Effects. Proceedings of the CIE Symposium on Light and Health: Non Visual Effects. 2005 Mar;p. 21-26. , Mar-2005